

MLOps Lab Experiment 3 – EC-1

Deploy and Serve the ML Model as a Microservice

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The contents below follow the Lab 3 EC-1 Deliverables Checklist.

1. /docs Page – Four Endpoints

The FastAPI documentation page shows the four required endpoints: GET /health, GET /model-info, POST /predict, and POST /predict-batch.

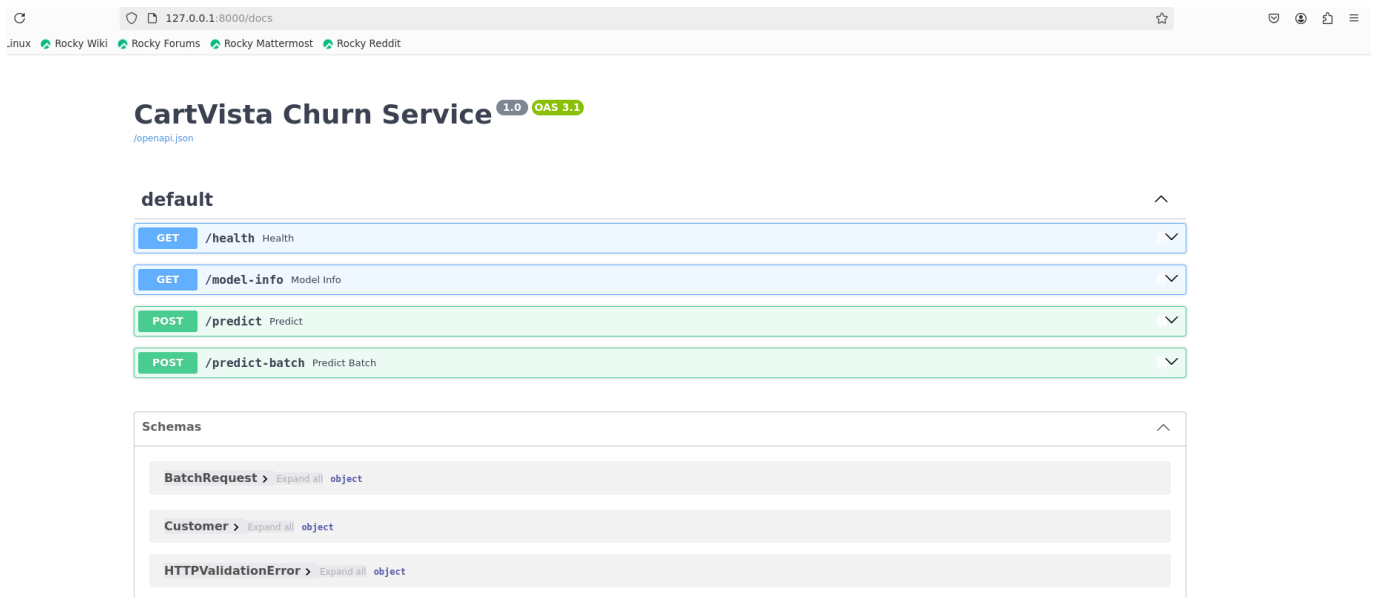


Figure 1. CartVista Churn Service FastAPI /docs page showing all four endpoints.

2. /predict – At-Risk, Loyal and Validation Rejection

The API validates customer inputs before prediction. Invalid values such as a negative tenure are rejected with HTTP 422.

```
[[20]-bash: $ \[20]-curl: command not found
(.venv) (base) s2025ae05230@ip-10-3-17-177:~/workspace/cartvista-mlops$
(.venv) (base) s2025ae05230@ip-10-3-17-177:~/workspace/cartvista-mlops$ curl -s -X POST http://127.0.0.1:8000/predict \
-H "Content-Type: application/json" \
-d '{"tenure_months": -5, "monthly spend": 2500, "orders_per month": 1,
    "support_tickets": 6, "uses_discounts": 0, "app_sessions": 2}'
{"detail":{"type":"greater_than_equal","loc":["body","tenure_months"],"msg":"Input should be greater than or equal to 0","input":-5,"ctx":{"ge":0}}}(.venv) (base) s2025ae05230
@ip-10-3-17-177:~/workspace/cartvista-mlops$
```

Figure 2. Terminal evidence of the /predict validation rejection: tenure_months = -5 returns HTTP 422.

```
@ip-10-3-17-177:~/workspace/cartvista-mlops$ python load_test.py
n=300 p50=5.7ms p95=7.4ms p99=9.1ms max=13.0ms
(.venv) (base) s2025ae05230@ip-10-3-17-177:~/workspace/cartvista-mlops$
```

Figure 3. Terminal evidence showing successful /predict requests from the service.

3. Load Test – Single vs Batch Economics (E2)

The load test compares 300 individual `/predict` calls with 3 `/predict-batch` calls containing 100 customers each. The actual measured result is **300 single calls: 1862.93 ms** and **3 batch calls: 28.05 ms**, giving a **66.43× speedup** for batching.

```
min 300 p50 3.7ms p99 7.1ms p99.9 9.1ms max 13.6ms
(.venv) (base) s2025ae05230@ip-10-3-17-177:~/workspace/cartvista-mlops$ python load_test.py
300 single calls : 1862.93 ms
3 batch calls    : 28.05 ms
Speedup         : 66.43x
(.venv) (base) s2025ae05230@ip-10-3-17-177:~/workspace/cartvista-mlops$
```

Figure 4. Terminal output for E2 showing 300 single calls, 3 batch calls, and 66.43× speedup.

E2 Explanation: The batch approach is much faster because 300 individual requests require 300 separate HTTP requests, JSON serialization/deserialization operations, request handling, and model-invocation overhead. With batching, the same 300 customers are processed using only 3 HTTP requests, with 100 customers per request. Therefore, most of the saved time comes from reducing per-request overhead rather than changing the ML model itself.

4. E1 – Probability Responses

E1 requires `churn_probability` values for one at-risk profile and one loyal profile using `predict_proba`. The supplied screenshots do not show those probability-valued responses, so no values are fabricated here.

5. E3 – Champion Hot-Swap

The service resolves the registry alias **`models:/cartvista-churn@champion`** during startup and stores the resolved model version and run ID in memory. Moving the `@champion` alias while the service is running therefore does not immediately change `/model-info`. Restarting the service causes the alias to be resolved again and the new champion version to be loaded.

E3 Answer: One risk is that a running service can continue serving an older model even after the registry's `@champion` alias has moved, creating a mismatch between the model intended for production and the model actually serving predictions.

6. Reflection

The service is fast and correct today. Two ways it could quietly become wrong next month without a single error appearing in its logs are: (1) the incoming customer data distribution could drift, making predictions less accurate while the API continues returning valid responses; and (2) the relationship between customer features and churn could change because customer behavior or business conditions change, causing concept drift. In both cases, the service can remain healthy and technically correct while its business accuracy deteriorates.

7. Deliverables Status

Required item	Evidence
/docs screenshot with four endpoints	Included
At-risk and loyal <code>/predict</code> calls + 422 rejection	Included
<code>load_test.py</code> percentile / load evidence	Included
E2 single-vs-batch comparison	Included – 66.43× speedup
E1 probability responses	Written requirement explained
E3 alias-resolution answer	Included
Reflection	Included